ENSET NLP / LAB 03

Chaotic Text Generator

Text generation has no single reference answer.
We compared a POS-HMM, a role-split HMM and a trigram with back-off.
We trained on 960 synthetic sentences and held out 240 by document.
We generated 500 sentences per model and temperature with seed 42.
We used Viterbi to recover POS tags and tested it on English EWT.

Synthetic generation at T = 1.0

Model	Valid pattern	Plausible pair	Novelty	Distinct-2
Plain HMM	16.2%	14.8%	99.8%	0.198
Role-split HMM	100.0%	60.6%	99.4%	0.148
Trigram	57.0%	76.0%	91.2%	0.105

The plain HMM pools transitions after subject, object and prepositional nouns. Splitting those states tracks sentence position and kept every sampled pattern within the synthetic grammar. Word emissions still depend on a single state, so a valid structure can contain an implausible subject and verb.

Temperature and examples

Writing toy: Use the role-split HMM at T = 1.5. Distinct-2 rises from 0.148 at T = 1.0 to 0.161 while valid patterns stay at 100%.

Autocomplete: Use the trigram at T = 0.5. It conditions on recent words and gives its highest valid-pattern rate (63.6%) among the tested temperatures.

Good: The children eat the bread at the park. The structure is valid, the action is plausible and the sentence is novel.
Meaning failure: The foxes lose by the school. The pattern is valid, but the subject-verb pair is absent from training. The role-split HMM samples each word independently given its state.

Viterbi on synthetic and real text

Corpus / model	Accuracy	Known	Unknown	Whole sent.
Synthetic / HMM	100.0%	100.0%	n/a	100.0%
EWT / most-frequent tag	86.2%	90.8%	28.6%	31.9%
EWT / HMM + Viterbi	88.1%	92.9%	28.2%	40.4%

English EWT has 10,359 training tag patterns versus 16 in the synthetic corpus. Unseen words account for 7.5% of EWT test tokens. With add-k smoothing, unseen words get larger emission probabilities under rare tags, contributing to errors such as PROPN decoded as X or INTJ.

Source: Universal Dependencies English EWT (CC BY-SA 4.0).